

Part 3: Solution Verification

Submit Your Solution

After analyzing the ciphertext and making substitutions in the workspace above, enter your final solution below:

 Copy from Scratchpad

Final Plaintext:

Enter the complete decrypted message

```
chapter 1 - down the rabbit hole: alice is bored sitting on the riverbank with her sister, when she notices a talking, clothed white rabbit with a pocket watch run past. she follows it down a rabbit hole when suddenly she falls a long way to a curious hall with many locked doors of all sizes. she finds a small key to a door too small for her to fit, but through which she sees an attractive garden. she then discovers a bottle labelled 'drink me', the contents of which cause her to shrink too small to reach the key. a cake with 'eat me' on it causes her to grow to such a tremendous size her head hits the ceiling.
```

Solution Key:

26 letters (a-z), each exactly once

```
xcdqrlpkwzoufteyahnvisgjbm
```

 Check Answer!

 **CORRECT!! Well done!**

Plaintext: chapter 1 - down the rabbit hole: alice is bored sitting on the riverbank with her sister, when she notices a talking, clothed white rabbit with a pocket watch run past. she follows it down a rabbit hole when suddenly she falls a long way to a curious hall with many locked doors of all sizes. she finds a small key to a door too small for her to fit, but through which she sees an attractive garden. she then discovers a bottle labelled 'drink me', the contents of which cause her to shrink too small to reach the key. a cake with 'eat me' on it causes her to grow to such a tremendous size her head hits the ceiling.

Key: xcdqrlpkwzoufteyahnvisgjbm

Part 4: Encryption Tool

Mono-Alphabetic Substitution Tool

Use this tool to encrypt/decrypt with your own key and plaintext:

Plaintext:

Your message to encrypt

```
defend the east wall of the castle
```

Substitution Key:

26-letter substitution alphabet

```
phqgiumeaylnofdxjkrvcstzwb
```

 Random Key

 Encrypt 



☐ Remove Punctuation

Ciphertext:

Encrypted result

```
giuifg cei iprc tpnn du cei qprcni
```

Mono-Alphabetic Substitution Cipher Analysis

Decrypt the cipher text using frequency analysis and substitution techniques. Tools are provided to assist your cryptanalysis.

 Next Ciphertext

 Reset Workspace

 Calculate Frequencies

Part 1: Ciphertext Analysis

Reference: English Letter Frequencies

Use this frequency table to guide your cryptanalysis:

a	b	c	d	e	f	g	h	i	j	k	l	m
8.167	1.49	2.782	4.253	12.702	2.228	2.015	6.094	6.966	0.153	0.772	4.025	2.406
n	o	p	q	r	s	t	u	v	w	x	y	z
6.749	7.507	1.929	0.095	5.987	6.327	9.056	2.758	0.978	2.360	0.150	1.974	0.074

Current Ciphertext:

The encrypted text to be analyzed

dkxyvrh 1 - qegt vkr hxccwv keur: xuwdr wn cehrq nwwvtp et vkr hwsrhcxto gvwk krh nwnvrh, gkrt nkr tevdrn x vxuowtp, duevkrq gkwr hxccwv gvwk x yedorv gxvdk hit yxnv. nkr leuuegn wv qegt x hxccwv keur gkrt niqqrub nkr lxuun x uetp gxb ve x dihwein kxuu gvwk fxtb uedorq qeehn el xuu nwmrn. nkr lwtqn x nfxuu orb ve x qeeh vee nfxuu leh krh ve lww, civ vkheipk gkwk nkr nrrn xt xvvhxdvwsr pxhqrt. nkr vkrt qwndesrhn x cevur uxcruurq 'qhwto fr', vkr detvrtvn el gkwk dxinr krh ve nkhwtv vee nfxuu ve hrxdk vkr orb. x dxor gvwk 'rxv fr' et wv dxinr krh ve pheg ve nidk x vhrftrqein nwmr krh krqx kwvn vkr druwvtp.

Ciphertext Frequency Analysis

Letter frequencies in the current ciphertext:

a	b	c	d	e	f	g	h	i	j	k	l	m
0.000	1.037	2.282	3.942	8.091	1.452	3.112	5.602	2.075	0.000	8.506	1.452	0.415
n	o	p	q	r	s	t	u	v	w	x	y	z
7.469	1.867	1.452	3.32	11.618	0.622	4.979	5.602	9.959	6.639	7.884	0.622	0.000

Part 2: Interactive Substitution Workspace

Substitution Scratchpad

Note: Cipher text is in *lower case*. When you replace characters, the final plaintext characters will be shown in *upper case* automatically.

Working Area:

Make substitutions and track your progress

Ckxyvrh 1 - qegt vkr hxccwv keur: xuwCr wn cehrq nwwvwtP et vkr hwsrhcxto gwvk krh nwnvrh, gkrt nkr tevwCrn x vxuowtp, Cuevkrq gkwvr hxccwv gwvk x yeCorv gxvCk hit yxnv. nkr leuegn wv qegt x hxccwv keur gkrt niqqtub nkr lXuun x uetp gxb ve x Cihwein kXuu gwvk fxtb ueCorq qeehn el xuu nwmrn. nkr lwtqn x nfxuu orb ve x qeeh vee nfxuu leh krh ve lwv, civ vkheipk gkwCk nkr nrrn xt xvvhxCvwsr pxhqrt. nkr vkrt qwnCesrhN x cevur uxruunq 'qhwtO fr', vkr Cetvrtvn el gkwCk Cxinr krh ve nkhwto vee nfxuu ve hrxCk vkr orb. x Cxor gwvk 'rxv fr' et wv Cxinrn krh ve pheg ve niCk x vhrftrqein nwmr krh krXq kwvn vkr CrwuwtP.

Substitution Tools

Replace cipher characters with plaintext:

This function is case *insensitive* and replaces cipher text (lower case) with plain text (upper case):

Replace cipher:

cipher character

with plaintext:

plain character

✓ Apply

Undo substitutions (case sensitive):

Use this function to undo unwanted changes by specifying exact characters:

Replace text:

any text

with text:

replacement

↺ Replace

Replacement History

You replaced d by C